

Matrix Form of the Kohn–Sham Equations for Small Basis Sets

Marco Fronzi

1 From Density Functional Theory to Kohn–Sham Orbitals

The central idea of Density Functional Theory (DFT) is that the ground-state properties of a many-electron system can be described using only the electron density:

$$n(\mathbf{r}). \tag{1}$$

Instead of solving directly for the complicated many-electron wavefunction

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N), \tag{2}$$

DFT states that the total energy can be written as a functional of the density:

$$E[n] = F[n] + \int V_{\text{ext}}(\mathbf{r}) n(\mathbf{r}) d\mathbf{r}, \tag{3}$$

where:

- $n(\mathbf{r})$ is the electron density,
- V_{ext} is the external ionic potential,
- $F[n]$ contains the kinetic energy and electron–electron interactions.

The problem is that the exact form of $F[n]$ is unknown.

To make the problem computationally tractable, Kohn and Sham introduced an auxiliary system of non-interacting electrons that reproduces exactly the same electron density as the real interacting system.

The key step is expressing the density in terms of single-particle orbitals:

$$n(\mathbf{r}) = \sum_i f_i |\psi_i(\mathbf{r})|^2, \tag{4}$$

where:

- $\psi_i(\mathbf{r})$ are the Kohn–Sham orbitals,
- f_i are occupation numbers.

For example, in a closed-shell system each occupied orbital contains two electrons of opposite spin, so $f_i = 2$.

This transformation is extremely important because solving for single-electron orbitals is much simpler than solving directly for the full many-electron wavefunction.

The total energy functional is rewritten as

$$E[n] = T_s[n] + \int V_{\text{ext}}(\mathbf{r}) n(\mathbf{r}) d\mathbf{r} + E_H[n] + E_{xc}[n], \tag{5}$$

where:

- $T_s[n]$ is the kinetic energy of non-interacting electrons,
- $E_H[n]$ is the Hartree electrostatic energy,
- $E_{xc}[n]$ is the exchange-correlation functional.

The Kohn–Sham orbitals are then obtained by minimizing the total energy with respect to the orbitals while preserving orbital orthonormality.

This variational procedure leads to the Kohn–Sham equations:

$$\hat{H}_{\text{KS}}\psi_i(\mathbf{r}) = \epsilon_i\psi_i(\mathbf{r}), \quad (6)$$

where

$$\hat{H}_{\text{KS}} = -\frac{1}{2}\nabla^2 + V_{\text{eff}}(\mathbf{r}), \quad (7)$$

and

$$V_{\text{eff}} = V_{\text{ext}} + V_H + V_{xc}. \quad (8)$$

The important conceptual point is that the complicated many-body physics is absorbed into the effective potential V_{eff} , allowing the problem to be solved as an effective single-particle system.

The Kohn–Sham orbitals themselves are not exact physical single-electron wavefunctions. Instead, they are auxiliary mathematical orbitals constructed so that their density reproduces the exact interacting electron density.

2 Kohn–Sham Density Functional Theory

In Kohn–Sham Density Functional Theory (DFT), the interacting many-electron problem is mapped onto an auxiliary system of non-interacting electrons moving in an effective potential. The Kohn–Sham equations are formally written as

$$\hat{H}_{\text{KS}}\psi_i(\mathbf{r}) = \epsilon_i\psi_i(\mathbf{r}), \quad (9)$$

where

$$\hat{H}_{\text{KS}} = -\frac{1}{2}\nabla^2 + V_{\text{eff}}(\mathbf{r}), \quad (10)$$

and

$$V_{\text{eff}} = V_{\text{ext}} + V_H + V_{xc}. \quad (11)$$

Here:

- V_{ext} is the external ionic potential,
- V_H is the Hartree electrostatic potential,
- V_{xc} is the exchange-correlation potential.

The Kohn–Sham orbitals are expanded using a finite basis set:

$$\psi_i(\mathbf{r}) = \sum_{\mu} c_{\mu i} \phi_{\mu}(\mathbf{r}), \quad (12)$$

where:

- $\phi_{\mu}(\mathbf{r})$ are basis functions,

- $c_{\mu i}$ are expansion coefficients.

This transforms the continuous differential equation into a finite-dimensional matrix eigenvalue problem.

The following sections illustrate this explicitly for basis sets containing 2 and 4 basis functions.

3 Example 1: Basis Set of 2 Functions

Consider two basis functions:

$$\phi_1(\mathbf{r}), \quad \phi_2(\mathbf{r}). \quad (13)$$

Each Kohn–Sham orbital can be written as

$$\psi_i(\mathbf{r}) = c_{1i}\phi_1(\mathbf{r}) + c_{2i}\phi_2(\mathbf{r}). \quad (14)$$

Substituting this expansion into the Kohn–Sham equation gives

$$\hat{H}_{\text{KS}}(c_{1i}\phi_1 + c_{2i}\phi_2) = \epsilon_i(c_{1i}\phi_1 + c_{2i}\phi_2). \quad (15)$$

Projecting onto each basis function and integrating over space produces the matrix form:

$$\sum_{\nu=1}^2 H_{\mu\nu} c_{\nu i} = \epsilon_i \sum_{\nu=1}^2 S_{\mu\nu} c_{\nu i}, \quad (16)$$

where

$$H_{\mu\nu} = \langle \phi_\mu | \hat{H}_{\text{KS}} | \phi_\nu \rangle \quad (17)$$

is the Hamiltonian matrix, and

$$S_{\mu\nu} = \langle \phi_\mu | \phi_\nu \rangle \quad (18)$$

is the overlap matrix.

Explicitly, the generalized eigenvalue problem becomes

$$\begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} \begin{pmatrix} c_{1i} \\ c_{2i} \end{pmatrix} = \epsilon_i \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} c_{1i} \\ c_{2i} \end{pmatrix}. \quad (19)$$

If the basis functions are orthonormal,

$$S_{\mu\nu} = \delta_{\mu\nu}, \quad (20)$$

then

$$\mathbf{S} = \mathbf{I}, \quad (21)$$

and the equation simplifies to

$$\mathbf{H}\mathbf{c}_i = \epsilon_i \mathbf{c}_i. \quad (22)$$

This is simply a standard matrix diagonalization problem.

3.1 Physical Interpretation

For a closed-shell system containing 4 electrons:

- two spatial Kohn–Sham orbitals exist,
- each orbital can host two electrons of opposite spin.

Therefore:

Orbital	Occupation
ψ_1	2 electrons
ψ_2	2 electrons

The electron density becomes

$$n(\mathbf{r}) = 2|\psi_1(\mathbf{r})|^2 + 2|\psi_2(\mathbf{r})|^2. \quad (23)$$

However, with only two basis functions:

- no virtual orbitals exist,
- the wavefunctions have extremely limited flexibility,
- excitation properties are poorly described.

4 Example 2: Basis Set of 4 Functions

Now consider a larger basis set:

$$\phi_1(\mathbf{r}), \phi_2(\mathbf{r}), \phi_3(\mathbf{r}), \phi_4(\mathbf{r}). \quad (24)$$

The Kohn–Sham orbitals are expanded as

$$\psi_i(\mathbf{r}) = \sum_{\mu=1}^4 c_{\mu i} \phi_{\mu}(\mathbf{r}). \quad (25)$$

Explicitly,

$$\psi_i(\mathbf{r}) = c_{1i}\phi_1 + c_{2i}\phi_2 + c_{3i}\phi_3 + c_{4i}\phi_4. \quad (26)$$

Projecting the Kohn–Sham equation onto the basis produces

$$\sum_{\nu=1}^4 H_{\mu\nu} c_{\nu i} = \epsilon_i \sum_{\nu=1}^4 S_{\mu\nu} c_{\nu i}. \quad (27)$$

The matrix form is

$$\begin{pmatrix} H_{11} & H_{12} & H_{13} & H_{14} \\ H_{21} & H_{22} & H_{23} & H_{24} \\ H_{31} & H_{32} & H_{33} & H_{34} \\ H_{41} & H_{42} & H_{43} & H_{44} \end{pmatrix} \begin{pmatrix} c_{1i} \\ c_{2i} \\ c_{3i} \\ c_{4i} \end{pmatrix} = \epsilon_i \begin{pmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{pmatrix} \begin{pmatrix} c_{1i} \\ c_{2i} \\ c_{3i} \\ c_{4i} \end{pmatrix}. \quad (28)$$

Or compactly:

$$\boxed{\mathbf{H}\mathbf{c}_i = \epsilon_i \mathbf{S}\mathbf{c}_i} \quad (29)$$

which is again a generalized eigenvalue problem.

4.1 Occupation of Orbitals

Diagonalization of the 4×4 Hamiltonian produces four Kohn–Sham orbitals:

$$\psi_1, \psi_2, \psi_3, \psi_4. \quad (30)$$

For a 4-electron closed-shell system:

Orbital	Occupation
ψ_1	2 electrons
ψ_2	2 electrons
ψ_3	0 electrons
ψ_4	0 electrons

The density becomes

$$n(\mathbf{r}) = 2|\psi_1(\mathbf{r})|^2 + 2|\psi_2(\mathbf{r})|^2. \quad (31)$$

The higher orbitals ψ_3 and ψ_4 are unoccupied virtual orbitals.

4.2 Advantages of Larger Basis Sets

Compared with the 2-function basis example, the 4-function basis provides:

- improved variational flexibility,
- better representation of bonding,
- virtual orbitals for excitations,
- more accurate densities,
- improved polarization effects,
- and better approximation of the true wavefunction.

Increasing the basis size systematically improves the accuracy of the Kohn–Sham representation by enlarging the accessible region of Hilbert space.

5 Conceptual Interpretation

The basis functions are not themselves physical electrons. Instead, they are mathematical building blocks used to represent the electronic orbitals.

The true continuous Kohn–Sham problem:

$$\hat{H}_{\text{KS}}\psi_i = \epsilon_i\psi_i \quad (32)$$

is transformed into the matrix problem

$$\mathbf{H}\mathbf{c}_i = \epsilon_i\mathbf{S}\mathbf{c}_i. \quad (33)$$

This transformation is one of the central computational ideas of modern electronic structure theory.

Larger basis sets provide greater flexibility and allow the calculated orbitals to more accurately approximate the exact many-electron electronic structure.